Zweitore

Formeln, etc

v0

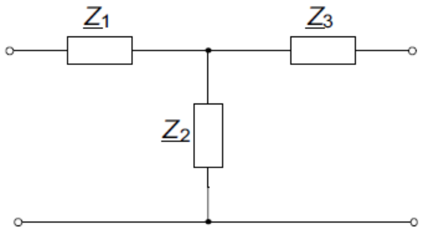
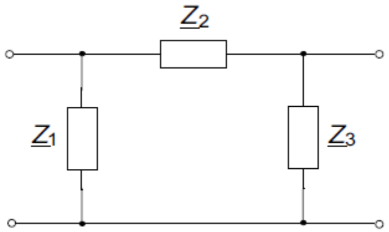
Quellen

[1] Führer, Arnold; Heidemann, Klaus; Nerreter, Wolfgang (2019): Grundgebiete der Elektrotechnik. Band 1: Stationäre Vorgänge: mit 426 Bildern, 71 durchgerechneten Beispielen, 65 Praxisbezügen und 145 Aufgaben mit Lösungen / Arnold Führer † (1940-2010), Klaus Heidemann, Wolfgang Nerreter. 10., neu bearbeitete Auflage. München: Hanser.

[2] Meier, Uwe; Stübbe, Oliver (2022): “Lineare Zweitore.” In: Elektrotechnik zum Selbststudium. Wiesbaden: Springer Fachmedien Wiesbaden, p. 335–368.

[3] “2.3: Scattering Parameters” (2020): Engineering LibreTexts. 2.3: Scattering Parameters. Available at: URL: https://eng.libretexts.org/Bookshelves/Electrical_Engineering/Electronics/Microwave_and_RF_Design_III_-_Networks_(Steer)/02%3A_Chapter_2/2.3%3A_Scattering_Parameters

T-Schaltung – Pi Schaltung

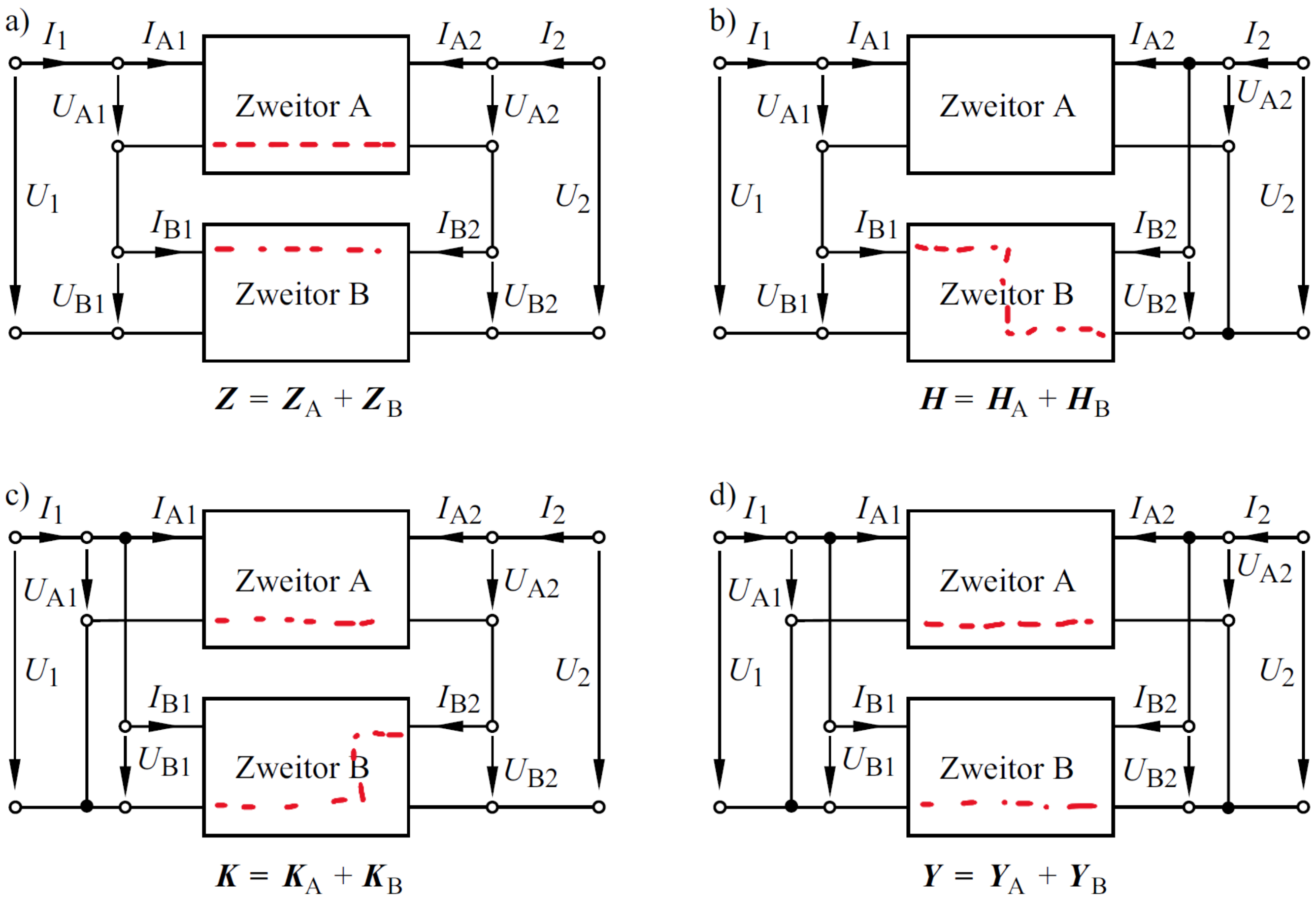
	T-Vierpol	π-Vierpol
		
($\underline{Y}$)	$\underline{Y}_{11} = \frac{\underline{Z}_2 + \underline{Z}_3}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$ $\underline{Y}_{12} = \frac{-\underline{Z}_2}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$ $\underline{Y}_{21} = \frac{-\underline{Z}_2}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$ $\underline{Y}_{22} = \frac{\underline{Z}_1 + \underline{Z}_2}{\underline{Z}_1 \underline{Z}_2 + \underline{Z}_1 \underline{Z}_3 + \underline{Z}_2 \underline{Z}_3}$	$\underline{Y}_{11} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3}$
($\underline{Z}$)	$\underline{Z}_{11} = \underline{Z}_1 + \underline{Z}_2$ $\underline{Z}_{12} = \underline{Z}_2$ $\underline{Z}_{21} = \underline{Z}_2$ $\underline{Z}_{22} = \underline{Z}_2 + \underline{Z}_3$	$\underline{Z}_{11} = \frac{\underline{Z}_1 (\underline{Z}_2 + \underline{Z}_3)}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$ $\underline{Z}_{12} = \frac{\underline{Z}_1 \underline{Z}_3}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$ $\underline{Z}_{21} = \frac{\underline{Z}_1 \underline{Z}_3}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$ $\underline{Z}_{22} = \frac{\underline{Z}_3 (\underline{Z}_1 + \underline{Z}_2)}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3}$

Umrechnungen

	$[Z]$	$[Y]$	$[A]$	$[H]$	$[K]$
$[Z]$	$\begin{matrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{matrix}$	$\begin{matrix} \frac{Y_{22}}{\det Y} & \frac{-Y_{12}}{\det Y} \\ \frac{-Y_{21}}{\det Y} & \frac{Y_{11}}{\det Y} \end{matrix}$	$\begin{matrix} \frac{A_{11}}{A_{21}} & \frac{\det A}{A_{21}} \\ \frac{1}{A_{21}} & \frac{A_{22}}{A_{21}} \end{matrix}$	$\begin{matrix} \frac{\det H}{H_{22}} & \frac{H_{12}}{H_{22}} \\ \frac{-H_{21}}{H_{22}} & \frac{1}{H_{22}} \end{matrix}$	$\begin{matrix} \frac{1}{K_{11}} & \frac{-K_{12}}{K_{11}} \\ \frac{K_{21}}{K_{11}} & \frac{\det K}{K_{11}} \end{matrix}$
$[Y]$	$\begin{matrix} \frac{Z_{22}}{\det Z} & \frac{-Z_{12}}{\det Z} \\ \frac{-Z_{21}}{\det Z} & \frac{Z_{11}}{\det Z} \end{matrix}$	$\begin{matrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{matrix}$	$\begin{matrix} \frac{A_{22}}{A_{12}} & \frac{-\det A}{A_{12}} \\ \frac{-1}{A_{12}} & \frac{A_{11}}{A_{12}} \end{matrix}$	$\begin{matrix} \frac{1}{H_{11}} & \frac{-H_{12}}{H_{11}} \\ \frac{H_{21}}{H_{11}} & \frac{\det H}{H_{11}} \end{matrix}$	$\begin{matrix} \frac{\det K}{K_{22}} & \frac{K_{12}}{K_{22}} \\ \frac{-K_{21}}{K_{22}} & \frac{1}{K_{22}} \end{matrix}$
$[A]$	$\begin{matrix} \frac{Z_{11}}{Z_{21}} & \frac{\det Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{matrix}$	$\begin{matrix} \frac{-Y_{22}}{Y_{21}} & \frac{-1}{Y_{21}} \\ \frac{-\det Y}{Y_{21}} & \frac{-Y_{11}}{Y_{21}} \end{matrix}$	$\begin{matrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{matrix}$	$\begin{matrix} \frac{-\det H}{H_{21}} & \frac{-H_{11}}{H_{21}} \\ \frac{-H_{22}}{H_{21}} & \frac{-1}{H_{21}} \end{matrix}$	$\begin{matrix} \frac{1}{K_{21}} & \frac{K_{22}}{K_{21}} \\ \frac{K_{11}}{K_{21}} & \frac{\det K}{K_{21}} \end{matrix}$
$[H]$	$\begin{matrix} \frac{\det Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ \frac{-Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{matrix}$	$\begin{matrix} \frac{1}{Y_{11}} & \frac{-Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\det Y}{Y_{11}} \end{matrix}$	$\begin{matrix} \frac{A_{12}}{A_{22}} & \frac{\det A}{A_{22}} \\ \frac{-1}{A_{22}} & \frac{A_{21}}{A_{22}} \end{matrix}$	$\begin{matrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{matrix}$	$\begin{matrix} \frac{K_{22}}{\det K} & \frac{-K_{12}}{\det K} \\ \frac{-K_{21}}{\det K} & \frac{K_{11}}{\det K} \end{matrix}$
$[K]$	$\begin{matrix} \frac{1}{Z_{11}} & \frac{-Z_{12}}{Z_{11}} \\ \frac{Z_{21}}{Z_{11}} & \frac{\det Z}{Z_{11}} \end{matrix}$	$\begin{matrix} \frac{\det Y}{Y_{22}} & \frac{Y_{12}}{Y_{22}} \\ \frac{-Y_{21}}{Y_{22}} & \frac{1}{Y_{22}} \end{matrix}$	$\begin{matrix} \frac{A_{21}}{A_{11}} & \frac{-\det A}{A_{11}} \\ \frac{1}{A_{11}} & \frac{A_{12}}{A_{11}} \end{matrix}$	$\begin{matrix} \frac{H_{22}}{\det H} & \frac{-H_{12}}{\det H} \\ \frac{-H_{21}}{\det H} & \frac{H_{11}}{\det H} \end{matrix}$	$\begin{matrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{matrix}$

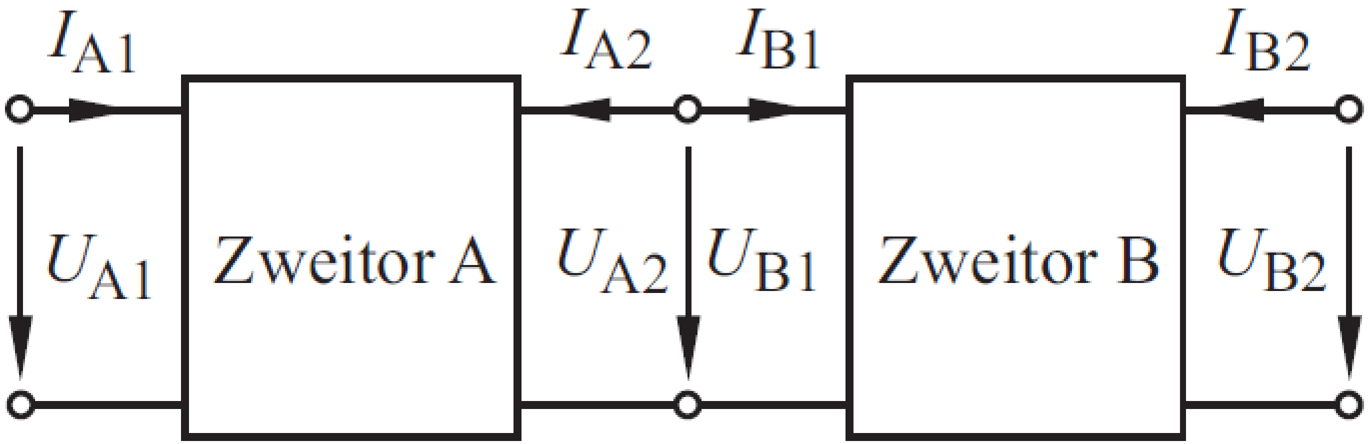
Quelle: [1]

Ersatzschaltungen



Quelle: [1]

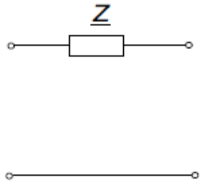
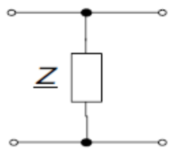
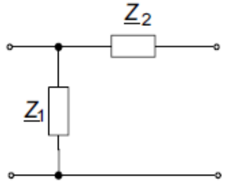
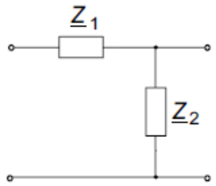
Kettenschaltung



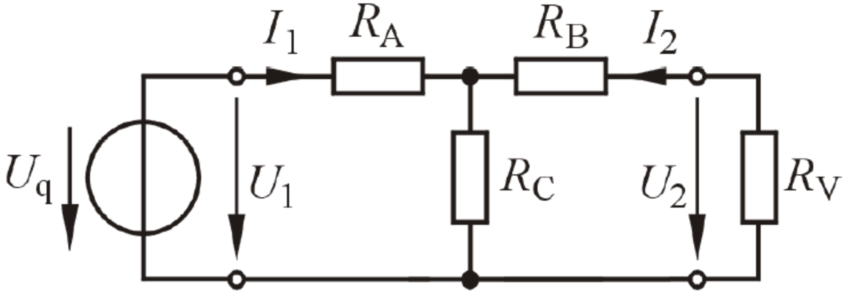
$$\boldsymbol{A} = \boldsymbol{A}_A \cdot \boldsymbol{A}_B$$

Torbedingung ist automatisch erfüllt

Elementarzweipole

	Längswiderstand	Querwiderstand	Γ-Vierpol 1	Γ-Vierpol 2
				
($\underline{Y}$)	$\underline{Y}_{11} = \frac{1}{\underline{Z}}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}}$	existieren nicht	$\underline{Y}_{11} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}_2}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}_2}$	$\underline{Y}_{11} = \frac{1}{\underline{Z}_1}$ $\underline{Y}_{12} = -\frac{1}{\underline{Z}_1}$ $\underline{Y}_{21} = -\frac{1}{\underline{Z}_1}$ $\underline{Y}_{22} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2}$
($\underline{Z}$)	existieren nicht	$\underline{Z}_{11} = \underline{Z}$ $\underline{Z}_{12} = \underline{Z}$ $\underline{Z}_{21} = \underline{Z}$ $\underline{Z}_{22} = \underline{Z}$	$\underline{Z}_{11} = \underline{Z}_1$ $\underline{Z}_{12} = \underline{Z}_1$ $\underline{Z}_{21} = \underline{Z}_1$ $\underline{Z}_{22} = \underline{Z}_1 + \underline{Z}_2$	$\underline{Z}_{11} = \underline{Z}_1 + \underline{Z}_2$ $\underline{Z}_{12} = \underline{Z}_2$ $\underline{Z}_{21} = \underline{Z}_2$ $\underline{Z}_{22} = \underline{Z}_2$
($\underline{A}$)	$\underline{A}_{11} = 1$ $\underline{A}_{12} = \underline{Z}$ $\underline{A}_{21} = 0$ $\underline{A}_{22} = 1$	$\underline{A}_{11} = 1$ $\underline{A}_{12} = 0$ $\underline{A}_{21} = \frac{1}{\underline{Z}}$ $\underline{A}_{22} = 1$	$\underline{A}_{11} = 1$ $\underline{A}_{12} = \underline{Z}_2$ $\underline{A}_{21} = \frac{1}{\underline{Z}_1}$ $\underline{A}_{22} = 1 + \frac{\underline{Z}_2}{\underline{Z}_1}$	$\underline{A}_{11} = 1 + \frac{\underline{Z}_1}{\underline{Z}_2}$ $\underline{A}_{12} = \underline{Z}_1$ $\underline{A}_{21} = \frac{1}{\underline{Z}_2}$ $\underline{A}_{22} = 1$

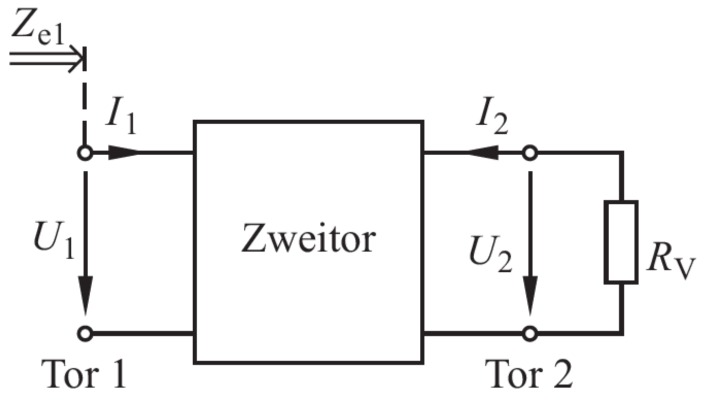
Übertragungsfaktoren

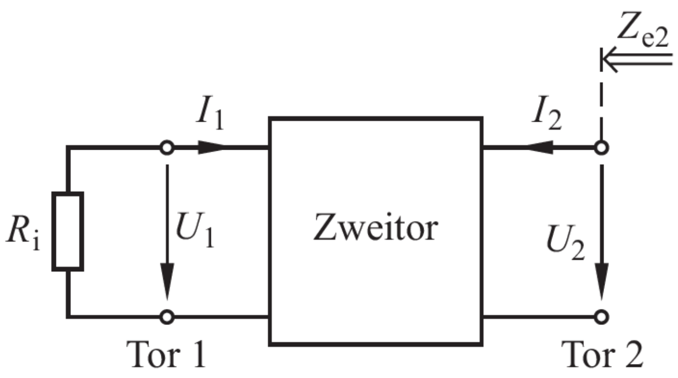


Z	$T_u = \dfrac{Z_{21}}{Z_{11} + G_V \cdot \det \boldsymbol{Z}}$	$T_i = \dfrac{-Z_{21}}{Z_{22} + R_V}$
Y	$T_u = \dfrac{-Y_{21}}{Y_{22} + G_V}$	$T_i = \dfrac{Y_{21}}{Y_{11} + R_V \cdot \det \boldsymbol{Y}}$
A	$T_u = \dfrac{1}{A_{11} + G_V A_{12}}$	$T_i = \dfrac{-1}{R_V A_{21} + A_{22}}$
H	$T_u = \dfrac{-H_{21}}{\det \boldsymbol{H} + G_V H_{11}}$	$T_i = \dfrac{H_{21}}{1 + R_V H_{22}}$
K	$T_u = \dfrac{K_{21}}{1 + G_V K_{22}}$	$T_i = \dfrac{-K_{21}}{\det \boldsymbol{K} + R_V K_{11}}$

Quelle: [1]

Ersatzwiderstände

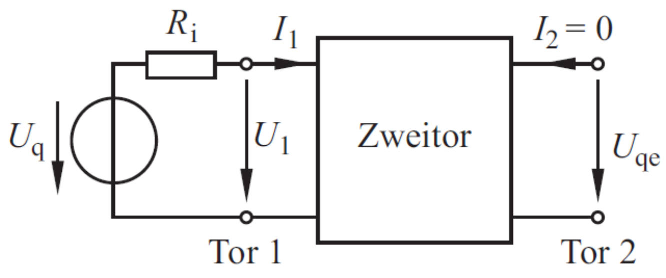




Z	$Z_{e1} = Z_{11} - \dfrac{Z_{12} Z_{21}}{Z_{22} + R_V}$	$Z_{e2} = Z_{22} - \dfrac{Z_{12} Z_{21}}{Z_{11} + R_i}$
Y	$Y_{e1} = Y_{11} - \dfrac{Y_{12} Y_{21}}{Y_{22} + G_V}$	$Y_{e2} = Y_{22} - \dfrac{Y_{12} Y_{21}}{Y_{11} + G_i}$
A	$Z_{e1} = \dfrac{A_{11} R_V + A_{12}}{A_{21} R_V + A_{22}}$	$Z_{e2} = \dfrac{A_{22} R_i + A_{12}}{A_{21} R_i + A_{11}}$
H	$Z_{e1} = H_{11} - \dfrac{H_{12} H_{21}}{H_{22} + G_V}$	$Y_{e2} = H_{22} - \dfrac{H_{12} H_{21}}{H_{11} + R_i}$
K	$Y_{e1} = K_{11} - \dfrac{K_{12} K_{21}}{K_{22} + R_V}$	$Z_{e2} = K_{22} - \dfrac{K_{12} K_{21}}{K_{11} + G_i}$

Quelle: [1]

Ersatzquelle



Z	$U_{qe} = \dfrac{Z_{21} U_q}{Z_{11} + R_i}$	$I_{qe} = \dfrac{-Y_{21} I_q}{Y_{11} + G_i}$	Y
A	$U_{qe} = \dfrac{U_q}{R_i \cdot A_{21} + A_{11}}$		
H	$I_{qe} = \dfrac{-H_{21} U_q}{H_{11} + R_i}$	$U_{qe} = \dfrac{K_{21} I_q}{K_{11} + G_i}$	K

Quelle: [1]

S-Parameter Umrechnung

	S	In terms of S
z	$Z'_{11} = z_{11} Z_0 \quad Z'_{12} = z_{12} Z_0$	$Z'_{21} = z_{21} Z_0 \quad Z'_{22} = z_{22} Z_0$
	$\delta_z = (1 + z_{11})(1 + z_{22}) - z_{12} z_{21}$ $S_{11} = [(z_{11} - 1)(z_{22} + 1) - z_{12} z_{21}]/\delta_z$ $S_{12} = 2z_{12}/\delta_z$ $S_{21} = 2z_{21}/\delta_z$ $S_{22} = [(z_{11} + 1)(z_{22} - 1) - z_{12} z_{21}]/\delta_z$	$\delta_S = (1 - S_{11})(1 - S_{22}) - S_{12} S_{21}$ $z_{11} = [(1 + S_{11})(1 - S_{22}) + S_{12} S_{21}]/\delta_S$ $z_{12} = 2S_{12}/\delta_S$ $z_{21} = 2S_{21}/\delta_S$ $z_{22} = [(1 - S_{11})(1 + S_{22}) + S_{12} S_{21}]/\delta_S$
y	$Y'_{11} = y_{11}/Z_0 \quad Y'_{12} = y_{12}/Z_0$	$Y'_{21} = y_{21}/Z_0 \quad Y'_{22} = y_{22}/Z_0$
	$\delta_y = (1 + y_{11})(1 + y_{22}) - y_{12} y_{21}$ $S_{11} = [(1 - y_{11})(1 + y_{22}) + y_{12} y_{21}]/\delta_y$ $S_{12} = -2y_{12}/\delta_y$ $S_{21} = -2y_{21}/\delta_y$ $S_{22} = [(1 + y_{11})(1 - y_{22}) + y_{12} y_{21}]/\delta_y$	$\delta_S = (1 + S_{11})(1 + S_{22}) - S_{12} S_{21}$ $y_{11} = [(1 - S_{11})(1 + S_{22}) + S_{12} S_{21}]/\delta_S$ $y_{12} = -2S_{12}/\delta_S$ $y_{21} = -2S_{21}/\delta_S$ $y_{22} = [(1 + S_{11})(1 - S_{22}) + S_{12} S_{21}]/\delta_S$
h	$H'_{11} = h_{11} Z_0 \quad H'_{12} = h_{12}$	$H'_{21} = h_{21} \quad H'_{22} = h_{22}/Z_0$
	$\delta_h = (1 + h_{11})(1 + h_{22}) - h_{12} h_{21}$ $S_{11} = [(h_{11} - 1)(h_{22} + 1) - h_{12} h_{21}]/\delta_h$ $S_{12} = 2h_{12}/\delta_h$ $S_{21} = -2h_{21}/\delta_h$ $S_{22} = [(1 + h_{11})(1 - h_{22}) + h_{12} h_{21}]/\delta_h$	$\delta_S = (1 - S_{11})(1 + S_{22}) + S_{12} S_{21}$ $h_{11} = [(1 + S_{11})(1 + S_{22}) - S_{12} S_{21}]/\delta_S$ $h_{12} = 2S_{12}/\delta_S$ $h_{21} = -2S_{21}/\delta_S$ $h_{22} = [(1 - S_{11})(1 - S_{22}) - S_{12} S_{21}]/\delta_S$

Quelle: [3]